Marzieh Allahyari

Disaster Resilience | Data Engineering, Analytics & AI
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Profile

Civil engineering Ph.D. researcher working across data engineering, data analytics, and AI to study disaster resilience. Experience cleaning and integrating longitudinal data, reconciling records across survey waves, and documenting data-quality decisions. Completed training in data integration and modern data warehousing, alongside data analysis and generative AI. Applies clustering and statistical modeling to examine housing recovery and demographic differences, and developed Recovery KPI Studio to support transparent analysis. Brings a background in structural reliability and uncertainty quantification.

Education

Ph.D., Civil and Environmental Engineering
University of Kansas 2024 – 2028 (expected)
Advisor: Dr. Elaina J. Sutley. Focus: data engineering, AI, and disaster resilience.

M.Sc., Structural Engineering
Tarbiat Modares University 2019 – 2022
Advisor: Prof. Masoud Soltani. Thesis: An approximate method to consider structural uncertainties in the response evaluation of reinforced concrete frames.

B.Sc., Civil Engineering
University of Tabriz 2015 – 2018

Research experience

Graduate Research Assistant
University of Kansas 2024 – present

- Doctoral research on long-term housing recovery after repeated flooding, using data from the NIST-funded Lumberton, North Carolina longitudinal field study.
- Use time-series clustering with Dynamic Time Warping (DTW) and latent class growth analysis (LCGA) to identify and compare household recovery pathways.
- Apply regression, regularized multinomial models, and linear mixed-effects models to examine recovery patterns and demographic differences across repeated observations.
- Evaluate analytical choices through sensitivity analysis and nonparametric group comparisons.

Master's Research and Projects, Structural Engineering
Tarbiat Modares University 2019 – 2022

- Developed Python tools for Monte Carlo simulation, first-order second-moment analysis, and dimension-reduction methods, and coupled them with the TaRa nonlinear finite-element program to automate large simulation studies of reinforced concrete members and frames.
- Finite-element modeling in Python; response-spectrum, time-history, and pushover analyses; steel and reinforced concrete design.

Selected software project

Recovery KPI Studio — Local analytical application built with Streamlit, pandas, and Plotly. Supports CSV/Excel imports, data-quality review, user-defined KPIs and equations, longitudinal exploration, interactive charts, and export of results and methodology.

Technical skills

Data engineering & business intelligence: SQL, Power BI, data cleaning, record reconciliation, multi-source integration, data-quality checks, and documentation; completed coursework in modern data warehousing.

Data analytics & statistics: Longitudinal data analysis, regression, regularized multinomial models, linear mixed-effects models, nonparametric tests, sensitivity analysis, and visualization.

AI & data science: Dynamic Time Warping and hierarchical clustering; latent class growth analysis; model selection and interpretation; working with repeated household observations.

Programming & software: Python: pandas, NumPy, SciPy, statsmodels, scikit-learn, matplotlib; Streamlit, Plotly, openpyxl, Excel, and Git.

Structural engineering & uncertainty: Monte Carlo simulation, first-order second-moment analysis, dimension-reduction methods, nonlinear finite-element analysis, TaRa, ETABS, SAFE, and OpenSees.

Publications & manuscripts

Under review. Allahyari, M., & Soltani, M. An efficient framework for uncertainty propagation in nonlinear reinforced concrete analysis using multiplicative dimension reduction and Laplace-transform maximum entropy. PLOS ONE.

Manuscript summary submitted. Allahyari, M., & Sutley, E. State-of-the-Art Longitudinal Field Studies in Disaster Housing Resilience and Recovery. Summary submitted to the Frontiers Research Topic Reviews in Resilience; editorial decision pending.

Additional manuscripts on long-term housing recovery are in preparation.

Teaching experience

Graduate Teaching Assistant, University of Kansas (Spring 2026). CE 260 Statics and Dynamics (Spring 2026): weekly office hours, problem-solving support, and course assistance for undergraduate engineering students.

Teaching Assistant, Tarbiat Modares University (2019 – 2022). Mathematics II and Probability & Statistics for undergraduate students. Tutored students in Python programming and the ETABS structural analysis software.

Awards & honors

- Society for Risk Analysis (SRA) Travel Award (2026) — travel support and complimentary registration
- Phi Kappa Phi Honor Society (2026) — nominated for membership, University of Kansas chapter
- Selected for the NHERI SPARC program (2025) at the Texas Advanced Computing Center (TACC), University of Texas at Austin — travel support offered by NHERI SimCenter through U.S. National Science Foundation funding
- Full scholarship for M.Sc. studies, Tarbiat Modares University (2019)
- Full scholarship for B.Sc. studies, University of Tabriz (2015)

Training & certificates

Building a Modern Data Warehouse - Data Engineering Bootcamp
Udemy · Completed Fall 2026.

Generative AI for data analytics
Udemy · Completed Summer 2026.

Data Analysis & AI: From Data to Intelligent Decisions 2026
Udemy · Completed Summer 2026.

Data Integration Fundamentals
Udemy · Completed Fall 2026.

Social & Behavioral Research
CITI Program, University of Kansas · Completed Summer 2026; valid through Summer 2029.